

APPLIED OPERATING SYSTEM



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MODULE 4

VIRTUAL MEMORY

OBJECTIVES

Upon completion of this module, you will be able to:

- Explain virtual memory and its benefits;
- Discuss the concept of demand paging and page replacement algorithms;
- Apply different page-replacement algorithms such as First-In First-Out (FIFO), Optimal Algorithm, and Least Recently Used (LRU) Algorithm in solving demand paging problems;



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VIRTUAL MEMORY OVERVIEW



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INTRODUCTION

The various memory management strategies that are used in computer systems have the same goal: **to keep many processes in memory simultaneously to allow multiprogramming.**

However, they tend to require the entire process to be in memory before the process can execute.



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WHAT IS VIRTUAL MEMORY?

Virtual Memory is a technique that allows the execution of processes that may not be completely in memory. One major advantage of this scheme is that programs can be larger than physical memory.

Further, virtual memory abstracts main memory into an extremely large, uniform array of storage, separating logical memory as viewed by the user from physical memory.



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VIRTUAL MEMORY BENEFITS

This technique frees programmers from the concerns of memory-storage limitations. Requiring that the entire process to be in main memory before they can be executed limits the size of a program to the size of physical memory.

However, an examination of real programs shows that, it need not be in main memory at the same time.

For example:

1. Programs often have code to handle unusual error conditions. Since these errors seldom, if ever, occur in practice, this code is almost never executed.



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VIRTUAL MEMORY BENEFITS

The ability to execute a program that is only partially in memory would confer many benefits:

1. A program would no longer be constrained by the amount of physical memory that is available. Users would be able to write programs for an extremely large virtual-address space.
2. Because each user program could take less physical memory, more programs could run at the same time.
3. Less I/O would be needed to load or swap each user program into memory, so each program would run faster.



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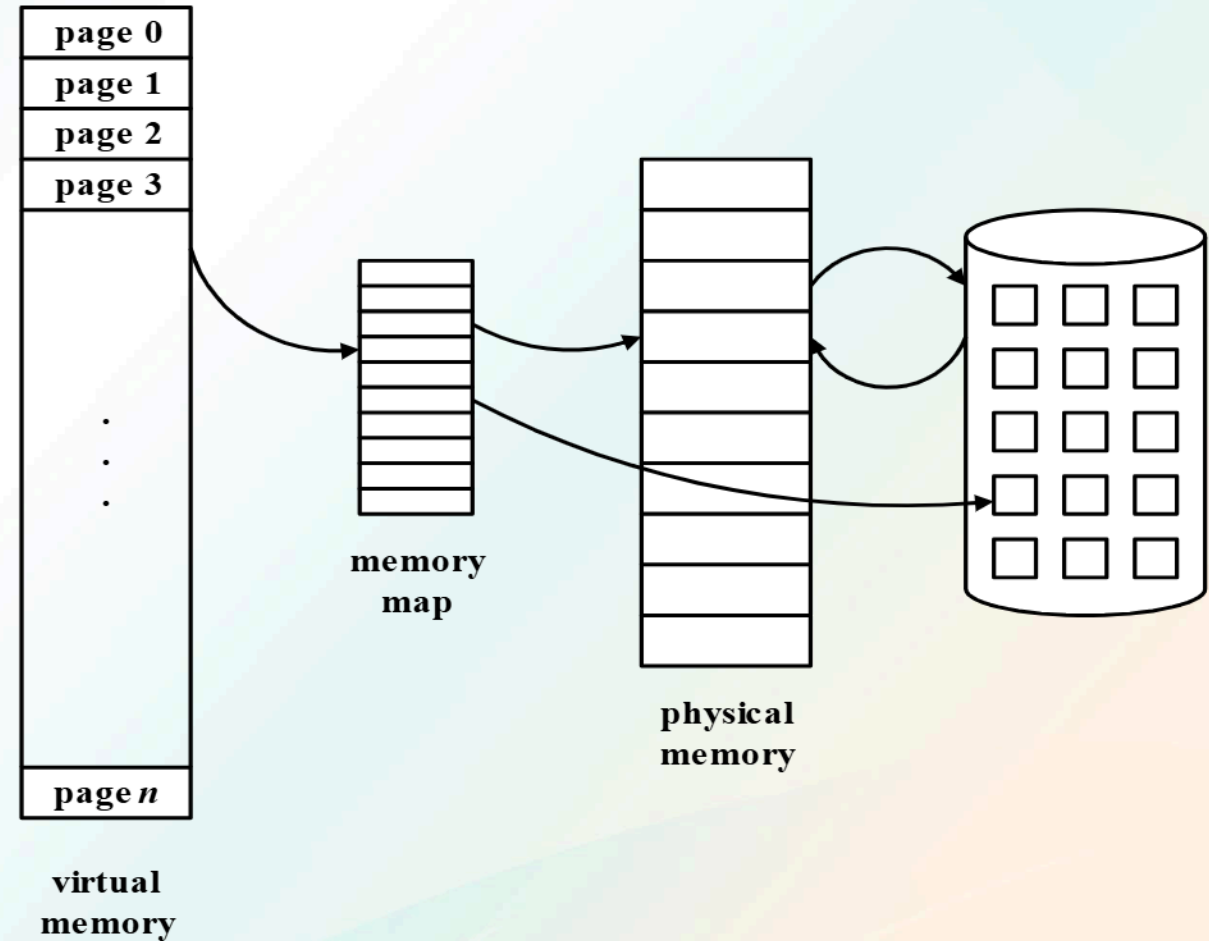


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VIRTUAL MEMORY

Virtual memory is the separation of user logical memory from physical memory. This separation allows programmers to have a very large virtual memory when only a small physical memory is available.



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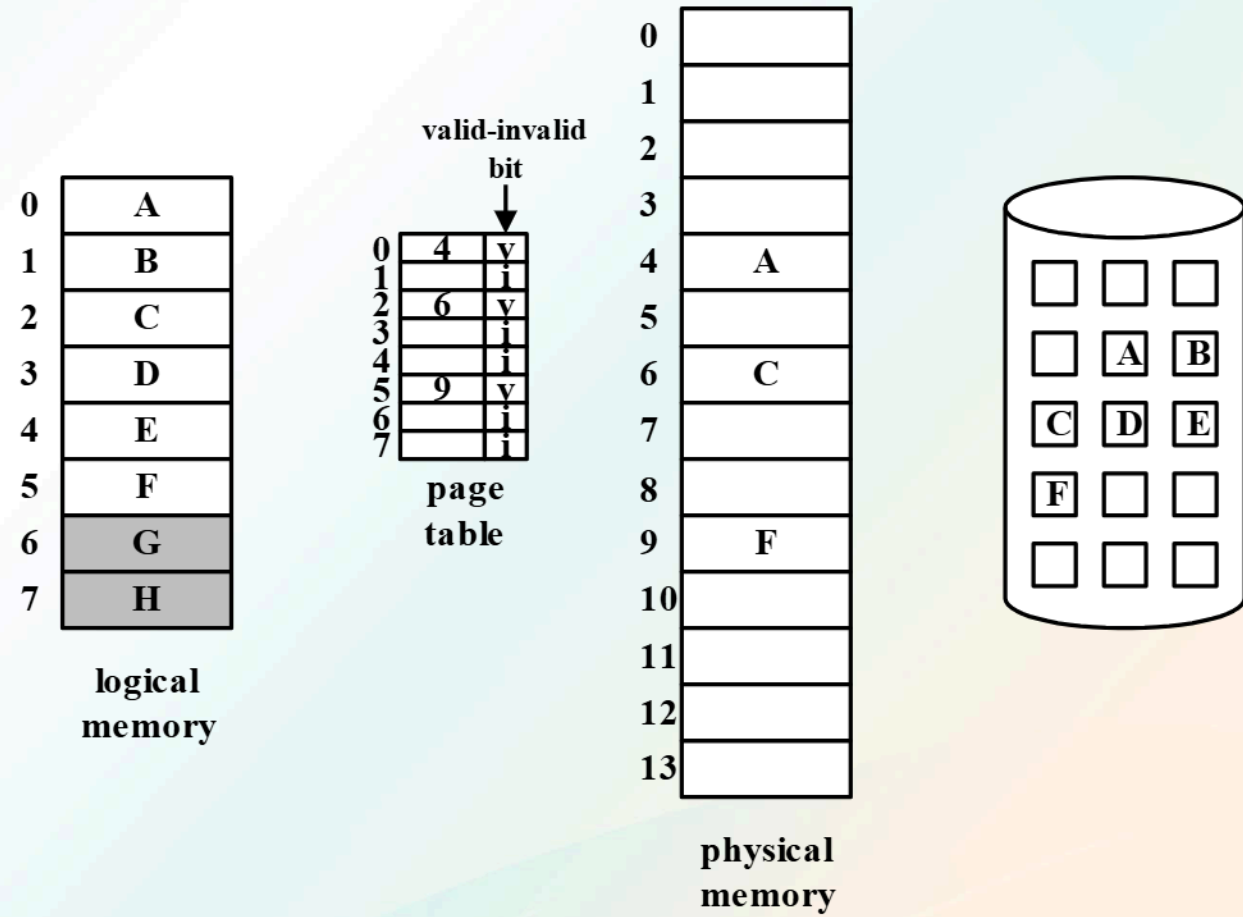


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DEMAND-PAGING

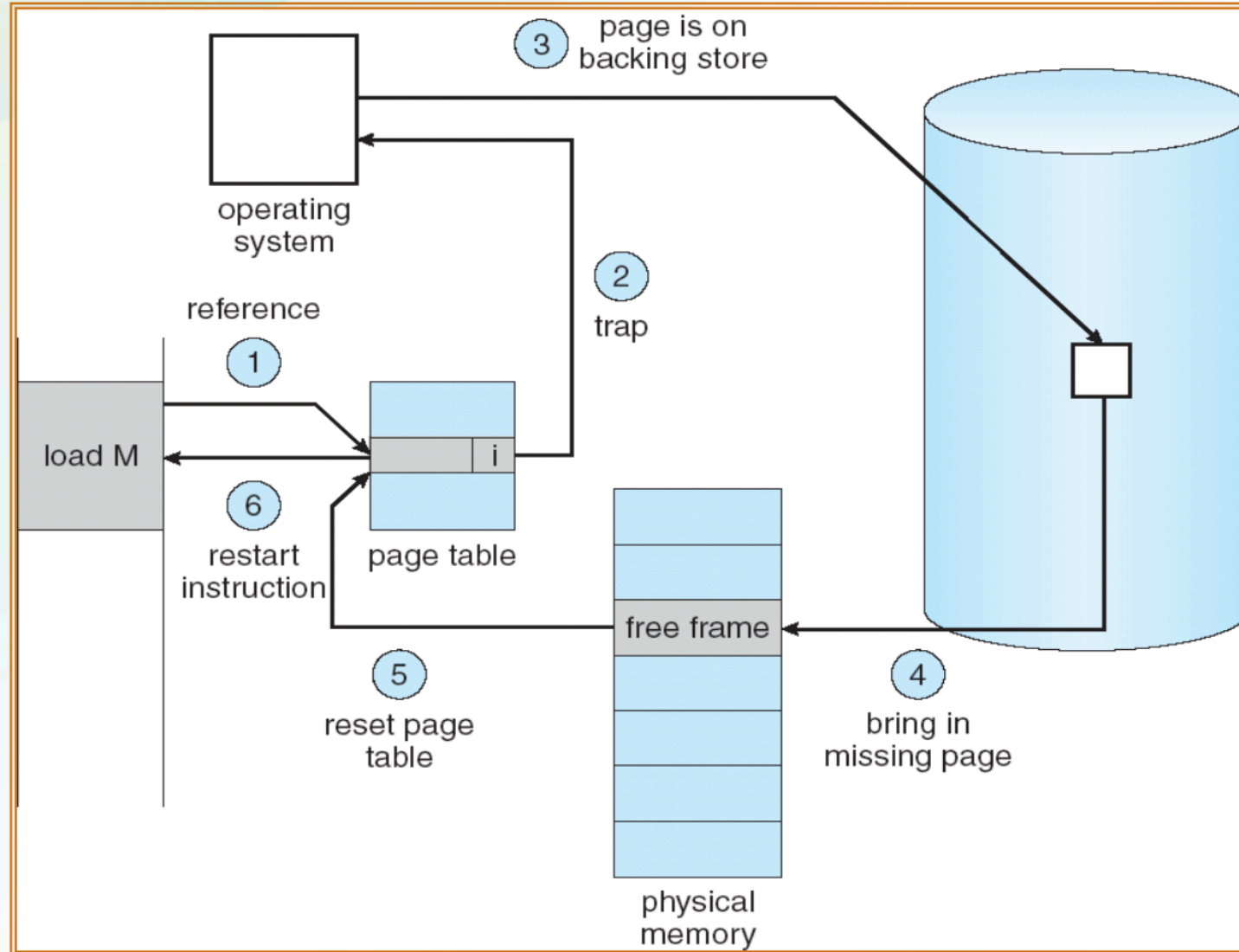
A **demand-paging system** is similar to a paging system with swapping. However, instead of swapping the entire process into memory, the OS (particular the pager) swaps only the necessary pages into memory (**lazy swapping**).



DEMAND-PAGING

- There is an additional bit in the page table which is the **valid-invalid bit**.
- This bit is set to **valid** to indicate that a corresponding **page is in memory**.
- This bit is set to **invalid** to indicate that the corresponding **page is in secondary storage**.
- If a process tries to use a page that is not in physical memory, then a **page-fault** will occur. This will cause a trap to the operating system indicating an invalid address error.

STEPS IN HANDLING PAGE-FAULT



DEMAND-PAGING

- In the extreme case, the system could start executing a process with no pages in memory. The process would immediately **fault** for the page with the **first instruction**.
- After the first page is brought into memory, the process would continue to execute, faulting as necessary until every page that it needed was actually in memory.
- This is **pure demand paging**: *never bring a page into memory until it is required.*



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DEMAND PAGING

- The **principle of locality of reference** ensures that programs do not access a new page of memory with each instruction execution.
- The **effectiveness of the demand paging** is based on a property of computer programs called the **locality of reference**.
- Analysis of programs shows that **most of their execution time is spent on routines** in which many instructions are executed repeatedly.
- These instructions may constitute a simple loop, nested loops, or a few procedures or functions that repeatedly call each other.



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DEMAND PAGING

- It is important to keep the page-fault rate **low** in a demand-paging system.
- Otherwise, the effective access time increases, slowing down process execution dramatically.
- A problem occurs if there is a need to transfer a page from disk to memory but there is no memory space available (*there are no free frames*). In other words, **memory is over-allocated**.



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PAGE-REPLACEMENT

- The operating system has several options at this point:
 - It could terminate the user process. However, demand paging is the operating system's attempt to improve the computer system's utilization and throughput.
 - Users should not be aware their processes are running on a paged system – paging should be logically transparent to the user.
 - Another possibility is **page replacement**. In this scheme, the operating system removes or replaces one of the existing pages in memory to give way for the incoming page.



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PAGE-REPLACEMENT

- A **page replacement algorithm** is necessary to select which among the pages currently residing in memory will be replaced.
- Page replacement takes the following approach:
 - If no frame is free, the system finds one that is currently being used and frees it.
 - Freeing a frame means transferring its contents to the disk and changing the page table (and all other tables) to indicate that the page is no longer in memory.



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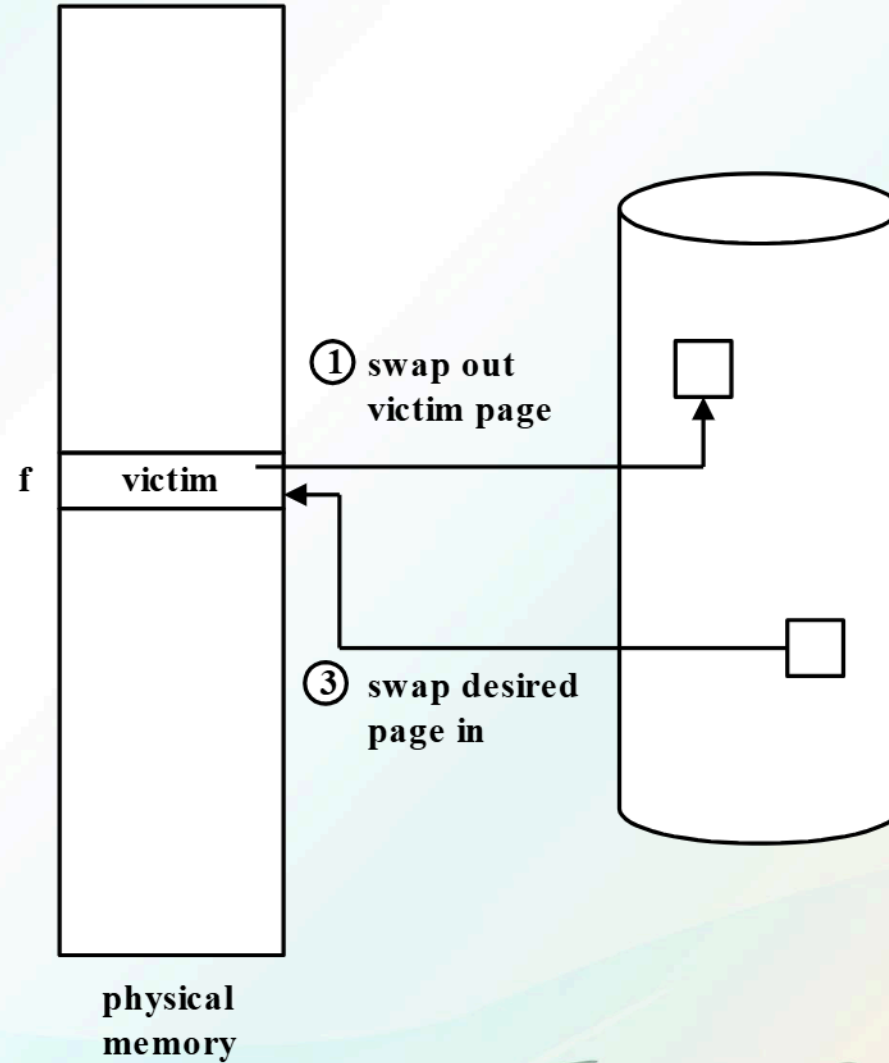
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PAGE-REPLACEMENT

f 0 i	
f	v

page table

- ② change to invalid
- ④ reset page table for new page



PAGE-REPLACEMENT

The page fault service routine is now modified to include page replacement:

1. Find the location of the desired page on the disk.
2. Find a free frame:
3. If there is a free frame, use it.
4. Otherwise, use a page-replacement algorithm to select a victim frame.
5. Write the victim page to the disk; change the page and frame tables accordingly.
6. Read the desired page into the (newly) free frame; change the page and frame tables.
7. Restart the user process.

PAGE-REPLACEMENT

- Notice that, if no frames are free, two page transfers (one out and one in) are required. This situation effectively doubles the page-fault service time and will increase the effective access time accordingly.
- To reduce this overhead, a **modify** or **dirty bit** is necessary for each page or frame.
- The **modify bit for a page is set** whenever any word or byte is written into, indicating that the **page has been modified**.



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PAGE-REPLACEMENT

- It is **no longer necessary to swap out pages** whose **modify bit is 0** (there was no modification). An **incoming page** may simply **overwrite an unchanged page**.
- There are **two major problems** in the implementation of demand paging:

- **Frame Allocation**

How many frames will the operating system allocate to a process, particularly if there are multiple processes?

- **Page Replacement**

How will the operating system select pages that are to be removed from memory to give way for incoming pages?



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PAGE REPLACEMENT TECHNIQUES



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PAGE-REPLACEMENT ALGORITHMS

- **Page replacement algorithms** are the techniques using which an OS decides which memory pages to swap out, write to disk when a page of memory needs to be allocated.
- A **good page-replacement algorithm** is one with a **low page-fault rate**.
- An algorithm is evaluated by running it on a particular string of memory references and computing the number of page faults. The string of memory references is called the **reference string**.
- Common Page-replacement algorithms or techniques are **First-In, First-Out (FIFO)**, **Optimal**, and **Least Recently Used (LRU)** Page Algorithm.



PAGE-REPLACEMENT ALGORITHMS

First-In First-Out (FIFO) Algorithm

This is the simplest page-replacement algorithm. When a page must be replaced, the oldest page is chosen.

Example:

Assume that there are **three frames** in memory and that the following is the **page reference string**:

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1



PAGE-REPLACEMENT ALGORITHMS

First-In First-Out (FIFO) Algorithm

		Reference string																			
Frame no.		7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1
	0	7	7	7	2	2	2	2	4	4	4	0	0	0	0	0	0	0	7	7	7
	1		0	0	0	0	3	3	3	2	2	2	2	2	1	1	1	1	1	0	0
	2			1	1	1	1	0	0	0	3	3	3	3	3	2	2	2	2	2	1

There are **15 page faults**.

■ *Page fault*



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PAGE-REPLACEMENT ALGORITHMS

First-In First-Out (FIFO) Algorithm

- It is not strictly necessary to record the time when a page is brought in. The OS simply creates a FIFO queue to hold all pages in memory. The page at the head of the queue is replaced if a frame is needed. When a page is brought into memory, it is inserted at the tail of the queue.
- The FIFO page-replacement algorithm is easy to understand and program. However, its performance is not always good. The page replaced may be an initialization module that was used a long time ago and is no longer needed.



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PAGE-REPLACEMENT ALGORITHMS

First-In First-Out (FIFO) Algorithm

- Notice that, even if algorithm selects for replacement a page that is in active use, everything still works correctly. After an active page is removed to bring in a new one, a fault occurs almost immediately to retrieve the active page.
- Thus, a bad replacement choice increases the page-fault rate and slow process execution, but does not cause incorrect execution.



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PAGE-REPLACEMENT ALGORITHMS

First-In First-Out (FIFO) Algorithm

- As a general rule, the **more frames available** in physical memory, the **lower the page-fault rate**.
- However, there are some instances where the page-fault rate may increase as the number of physical memory frames increases. This is known as **Belady's Anomaly**.



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PAGE-REPLACEMENT ALGORITHMS

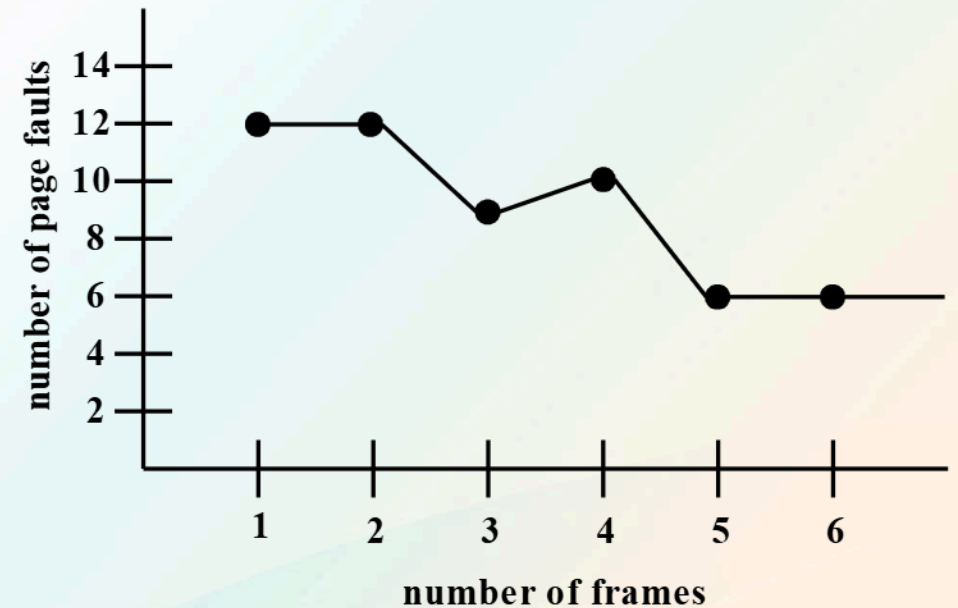
First-In First-Out (FIFO) Algorithm suffers from **Belady's Anomaly**.

Example:

Consider the reference string: **1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5**

Number of Frames	Page Faults
------------------	-------------

1	12
2	12
3	9
4	10
5	5
6	5



PAGE-REPLACEMENT ALGORITHMS

Optimal Algorithm

- This algorithm has the **lowest page-fault** rate of all algorithms.
- In an optimal algorithm, the page that will be replaced is the one that **will not be used for the longest period of time**.

		Reference string																			
Frame no.		7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1
	0	7	7	7	2	2	2	2	2	2	2	2	2	2	2	2	2	2	7	7	7
	1		0	0	0	0	0	0	4	4	4	0	0	0	0	0	0	0	0	0	0
	2			1	1	1	3	3	3	3	3	3	3	3	1	1	1	1	1	1	1

There are **9 page faults**.



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PAGE-REPLACEMENT ALGORITHMS

Optimal Algorithm

- In the example, the **optimal** page-replacement algorithm produced only **9 page faults** as compared to the **FIFO** algorithm which produced **15 page faults**.
- If the first three page faults were ignored (since all algorithms suffer from these initialization page faults), then the optimal algorithm is twice as good as FIFO.
- Unfortunately, this algorithm is **difficult to implement**, since it requires **future knowledge** of the reference string. The optimal algorithm is used mainly for comparison studies.



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PAGE-REPLACEMENT ALGORITHMS

Least Recently Used (LRU) Algorithm

- This algorithm uses the **recent past** to approximate the near future. It simply replaces the page that **has not been used for the longest period of time**.

		Reference string																			
Frame no.	0	7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1
	1	7	7	7	2	2	2	2	4	4	4	0	0	0	1	1	1	1	1	1	1
	2		0	0	0	0	0	0	0	0	3	3	3	3	3	3	0	0	0	0	0
	3			1	1	1	3	3	3	2	2	2	2	2	2	2	2	2	7	7	7

 Page Hit

There are **12 page faults**.

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PAGE-REPLACEMENT ALGORITHMS

Least Recently Used (LRU) Algorithm

- The LRU algorithm produced **12 page faults**. Although it is not as good as the **9** of the **optimal algorithm** (in fact, no other algorithm will produce less than 9 page faults for this example), it is much better than the **15** of the **FIFO algorithm**.
- The LRU policy is often used as a page replacement algorithm and is considered to be quite good. The major problem is how to implement LRU replacement.



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REFERENCES

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- **Tomsho, G. (2019). Guide to Operating Systems, Cengage Learning.**



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